

TVER, Russian Federation

WMO: 274020

Lat: 56.830N

Lon: 35.752E

Elev: 478

StdP: 14.44

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-6.8	-17.1	2.2	-12.7	-11.2	3.1	-6.5	19.8	27.5	17.8	25.9	1.8	100	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	85.6	67.1	81.9	66.2	78.5	64.7	69.7	80.8	68.0	78.2	66.4	75.9	6.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	97.3	73.9	64.4	92.2	72.4	62.8	87.1	70.5	34.0	81.0	32.6	78.7	31.3	76.5	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.6	15.7	13.4	DB	-20.4	89.3	5.8	4.2	-24.6	92.3	-28.0	94.8	-31.2	97.2	-35.4	100.2	
			WB	-20.6	72.5	5.8	2.9	-24.8	74.7	-28.1	76.4	-31.4	78.0	-35.6	80.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.8	18.3	18.9	28.5	42.9	54.3	61.7	66.9	62.3	52.1	41.4	30.4	22.7
	DBStd	19.26	12.59	12.82	8.79	8.45	8.74	6.78	5.94	6.42	6.62	7.59	9.63	11.96
	HDD50	4590	981	870	666	246	55	1	0	1	53	280	590	847
	HDD65	8668	1446	1290	1131	663	347	143	48	129	389	731	1039	1312
	CDD50	1609	0	0	0	33	188	353	523	383	115	14	0	0
	CDD65	214	0	0	0	0	15	46	106	46	1	0	0	0
	CDH74	1956	0	0	0	6	168	414	921	432	15	0	0	0
	CDH80	535	0	0	0	0	44	102	266	123	0	0	0	0
Wind	WSAvg	5.1	5.5	5.5	5.7	5.4	5.5	4.8	4.2	4.1	4.4	5.3	5.6	5.6
Precipitation	PrecAvg	26.2	1.7	1.5	1.3	1.3	2.5	3.0	3.1	3.0	2.4	2.6	2.0	1.7
	PrecMax	36.4	3.0	2.8	2.2	2.6	6.8	5.7	8.9	6.9	5.9	6.4	4.3	3.1
	PrecMin	17.6	0.4	0.4	0.3	0.3	0.9	0.2	0.1	0.3	1.2	0.7	0.1	0.7
	PrecStd	5.0	0.6	0.6	0.5	0.6	1.3	1.3	2.1	1.6	1.1	1.4	1.0	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.0	40.1	54.5	73.6	85.4	87.3	92.1	89.5	76.7	64.9	50.6	45.6
		MCWB	38.5	37.3	45.1	56.3	65.6	67.3	70.1	68.4	63.4	56.0	48.1	44.1
	2%	DB	36.4	37.4	47.2	68.2	78.5	82.9	86.4	82.7	71.0	58.4	46.8	41.4
		MCWB	34.9	34.9	39.8	52.5	62.2	66.1	67.9	66.6	60.3	52.8	44.7	40.2
	5%	DB	34.7	35.8	42.8	62.8	73.6	78.9	82.5	78.6	66.6	54.7	44.3	38.1
		MCWB	33.4	33.6	37.5	50.2	59.1	64.2	66.9	65.1	58.2	51.1	42.5	36.6
	10%	DB	33.3	34.2	39.5	58.2	69.3	74.9	79.0	74.5	62.9	51.9	41.7	35.6
		MCWB	32.2	32.6	35.5	48.1	56.7	62.7	65.6	63.2	56.8	49.1	40.3	34.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.8	38.2	45.7	57.3	68.6	70.4	72.8	71.0	65.3	57.5	48.9	44.0
		MCDB	39.6	39.4	53.7	71.6	82.4	82.1	84.5	82.9	72.8	61.6	50.2	44.8
	2%	WB	35.2	35.5	40.7	54.0	63.8	68.3	70.1	68.5	61.9	53.9	45.1	40.3
		MCDB	36.1	36.7	45.1	64.1	74.6	79.2	81.3	78.9	67.9	57.3	46.5	41.4
	5%	WB	33.7	34.1	38.1	51.5	61.2	66.3	68.5	66.4	59.6	51.6	42.7	36.8
		MCDB	34.5	35.3	41.6	60.6	70.7	75.4	78.7	76.0	65.1	54.5	43.9	37.8
	10%	WB	32.2	32.8	36.0	49.0	58.5	64.0	67.0	64.4	57.4	49.5	40.4	34.6
		MCDB	33.1	34.1	38.9	57.6	66.9	72.0	76.5	72.0	62.3	51.7	41.6	35.4
Mean Daily Temperature Range	5% DB	MDBR	9.3	11.7	13.7	17.4	18.6	17.2	17.9	17.1	15.2	10.7	7.5	7.8
		MCDBR	6.3	8.3	17.3	25.0	24.6	23.3	23.8	23.9	21.5	15.5	9.3	6.3
		MCWBR	6.5	7.0	11.6	12.6	11.0	10.4	9.3	9.8	11.6	10.2	8.2	6.2
	5% WB	MCDBR	6.3	7.6	14.5	21.7	22.2	20.1	20.2	20.6	18.5	13.3	8.5	6.1
		MCWBR	6.6	7.0	10.2	12.0	10.9	10.4	9.4	9.5	10.9	9.5	8.0	6.0
Clear-Sky Solar Irradiance	taub	0.283	0.322	0.341	0.397	0.386	0.371	0.408	0.411	0.369	0.346	0.317	0.278	
	taud	2.116	2.073	2.148	2.192	2.336	2.398	2.398	2.311	2.292	2.384	2.373	2.240	2.125
	Ebn at Noon	182	222	254	258	270	276	261	252	246	219	170	150	
	Edh at Noon	18	29	35	39	36	35	37	36	28	22	16	14	
All-Sky Solar Radiation	RadAvg	159	417	894	1262	1655	1727	1691	1337	849	382	139	85	
	RadStd	30	63	104	125	141	164	189	132	115	72	24	23	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-437	N/A	N/A
Regional (9 neighbors)		+1.24	N/A	N/A	N/A	N/A	N/A	-275	-401	N/A	N/A

Nomenclature: See separate page